of force is always in the direction of acceleration.

2) Linear momentum $\vec{P} = m\vec{v}$. Here $\vec{v}$ is the velocity. The direction of linear momentum is also in the direction of velocity.

3) Force $\vec{F} = q\vec{E}$, Here the electric charge 'q' is a scalar, and $\vec{E}$ is the electric field. Since charge can be positive or negative, the direction of force $\vec{F}$ is correspondingly either in the direction of $\vec{E}$ or opposite to the direction of $\vec{E}$.

2.5.1 Scalar Product of Two Vectors

Definition

The scalar product (or dot product) of two vectors is defined as the product of the magnitudes of both the vectors and the cosine of the angle between them.

Thus if there are two vectors $\vec{A}$ and $\vec{B}$ having an angle θ between them, then their scalar product is defined as $\vec{A} \cdot \vec{B} = AB\cos\theta$. Here, A and B are magnitudes of $\vec{A}$ and $\vec{B}$.

Properties

(i) The product quantity $\vec{A} \cdot \vec{B}$ is always a scalar. It is positive if the angle between the vectors is acute (i.e., θ < 90°) and negative if the angle between them is obtuse (i.e. 90°<θ< 180°).

(ii) The scalar product is commutative, i.e. $\vec{A}.\vec{B} = \vec{B}.\vec{A}$

(iii) The vectors obey distributive law i.e. $\vec{A}.\left(\vec{B} + \vec{C}\right) = \vec{A}.\vec{B} + \vec{A}.\vec{C}$

(iv) The angle between the vectors

$$\theta = \cos^{-1}\left[\frac{\vec{A} \cdot \vec{B}}{AB}\right]$$

(v) The scalar product of two vectors will be maximum when cos θ = 1, i.e. θ = 0°, i.e., when the vectors are parallel;

$$(\vec{A} \cdot \vec{B})_{max} = AB$$

(vi) The scalar product of two vectors will be minimum, when cos θ = –1, i.e. θ = 180°

$(\vec{A} \cdot \vec{B})_{min} = -AB$, when the vectors are anti-parallel.

(vii) If two vectors $\vec{A}$ and $\vec{B}$ are perpendicular to each other then their scalar product $\vec{A} \cdot \vec{B} = 0$, because cos 90°=0. Then the vectors $\vec{A}$ and $\vec{B}$ are said to be mutually orthogonal.

(viii) The scalar product of a vector with itself is termed as self–dot product and is given by $\left(\vec{A}\right)^2 = \vec{A} \cdot \vec{A} = AA\cos\theta = A^2$. Here angle $\theta = 0°$

The magnitude or norm of the vector $\vec{A}$ is $|\vec{A}| = A = \sqrt{\vec{A} \cdot \vec{A}}$

(ix) In case of a unit vector $\hat{n}$

$\hat{n} \cdot \hat{n} = 1 \times 1 \times \cos 0 = 1$. For example, $\hat{i}.\hat{i} = \hat{j}.\hat{j} = \hat{k}.\hat{k} = 1$

(x) In the case of orthogonal unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$,

$$\hat{i}.\hat{j} = \hat{j}.\hat{k} = \hat{k}.\hat{i} = 1 \cdot 1\cos 90° = 0$$

(xi) In terms of components the scalar product of $\vec{A}$ and $\vec{B}$ can be written as

$$\vec{A} \cdot \vec{B} = (A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) \cdot \left(B_x\hat{i} + B_y\hat{j} + B_z\hat{k}\right)$$

$= A_xB_x + A_yB_y + A_zB_z$, with all other terms zero.

The magnitude of vector $|\vec{A}|$ is given by

$$|\vec{A}| = A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

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